Computer Science and Engineering Department, SVNIT, Surat
BTech I CSE
Introduction to Computer Science (CS101)
Mid Term Syllabus

Ch-1-INTRODUCTION TO COMPUTER AND ITS ARCHITECTURE

Introduction and Characteristics, Computer Architecture, Generations, Classifications, Applications, Central Processing Unit and Memory, Communication between various Units, Processor Speed, Multiprocessor System, Peripheral Buses, Motherboard Demonstration.

Ch-2-NUMBER SYSTEMS

Introduction and type of Number System, Conversion between Number System, Arithmetic Operations in different Number System, Signed and Unsigned Number System.

Ch-3-COMPUTATIONAL PROBLEM SOLVING

Program Development Cycle, Pseudocode, Flowchart, Representing Information as Bits, Binary System, Storing Integers, Storing Fractions, Examples of Computational Problems, Iterative and Recursive Approaches to Solve Computational Problems.

Ch-4-MEMORY AND VARIOUS INPUT AND OUTPUT DEVICES

Introduction to Memory, Input and Output Devices, Memory Hierarchy, Primary Memory and its Types, Secondary Memory, Classification of Secondary Memory, Various Secondary Storage Devices, and their Functioning.

Ch-5-INTRODUCTION TO SYSTEM SOFTWARES AND PROGRAMMING LANGUAGES

Classification of Computer Languages, Introduction of Operating System, Evolution, Type and Function of OS, Unix Commands, Evolution and Classification of programming Language, Feature and Selection of good Programming Language, Development of Program, Algorithm and Flowchart, Program Testing and Debugging, Program Documentation and Paradigms, Characteristics of good Program.

Ch-6-WINDOWS OPERATING SYSTEM AND ITS ENVIRONMENT

Introduction to GUI based OS, Configuration, Setup, Services, Network Configuration.